

KOKOMO MUNICIPAL, IN, USA

WMO: 724387

Lat: 40.528N

Lon: 86.059W

Elev: 253

StdP: 98.32

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 54807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.4	(c) -14.0	(d) -22.4	(e) 0.5	(f) -16.5	(g) -19.1	(h) 0.7	(i) -13.2	(j) 14.3	(k) -2.3	(l) 12.8	(m) -1.7	(n) 4.1	(o) 280	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 32.7	(d) 23.7	(e) 31.6	(f) 23.2	(g) 30.2	(h) 22.5	(i) 25.3	(j) 30.9	(k) 24.5	(l) 29.5	(m) 23.6	(n) 28.3	(o) 5.0	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	19.0	28.6	22.7	18.0	27.5	22.2	17.4	26.9	78.3	30.8	75.3	29.9	71.7	28.0	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.9	(b) 10.6	(c) 9.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB													
			WB	-19.8	26.5	6.3	1.3	-24.3	27.4	-28.0	28.2	-31.5	28.9	-36.1	29.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-2.4	-1.7	5.2	11.1	17.6	22.0	23.4	22.4	19.5	12.7	5.8	0.3
	DBStd	10.54	6.71	6.57	6.14	4.93	4.80	3.35	3.01	2.75	3.80	4.97	5.44	5.53
	HDD10.0	1416	385	331	176	43	3	0	0	0	0	28	147	303
	HDD18.3	3074	641	562	411	221	73	8	2	3	32	187	376	560
	CDD10.0	1925	2	3	26	77	239	359	416	384	284	112	22	2
	CDD18.3	538	0	0	2	4	51	117	159	128	65	12	1	0
	CDH23.3	4820	0	0	18	56	460	992	1487	1074	620	111	2	0
	CDH26.7	1578	0	0	3	4	139	333	541	327	205	27	0	0
(14) Wind	WSAvg	4.3	5.4	5.1	5.0	5.2	4.5	3.8	3.1	2.8	3.1	4.1	4.8	5.0
(15) Precipitation	PrecAvg	1112	67	58	75	104	113	131	123	106	97	81	83	74
	PrecMax	1528	177	146	149	254	196	372	355	201	209	179	167	170
	PrecMin	766	22	11	18	27	35	29	38	26	23	25	20	21
	PrecStd	184	32	35	31	51	40	66	65	42	44	44	33	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	17.2	26.4	27.5	32.9	33.8	35.0	32.8	33.0	29.8	22.6	16.1
		MCWB	13.6	12.4	17.6	16.8	21.4	23.1	25.1	24.5	23.1	21.9	15.3	13.2
	2%	DB	12.4	13.6	20.9	25.2	30.0	32.3	32.7	31.3	31.3	27.0	18.9	12.9
		MCWB	10.5	11.4	15.0	15.9	20.6	22.9	24.5	24.0	22.5	19.1	14.8	10.9
	5%	DB	9.7	10.2	17.4	22.8	27.9	30.9	31.3	29.9	29.0	23.7	17.0	10.4
		MCWB	8.4	8.2	12.6	15.0	19.5	22.3	23.7	23.2	21.3	17.6	13.1	8.6
	10%	DB	6.9	7.1	14.2	20.7	26.3	28.6	29.7	28.2	27.3	21.3	14.1	7.6
		MCWB	4.6	4.8	10.9	14.4	18.9	21.0	22.8	22.2	19.4	16.5	10.5	5.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.8	17.9	19.0	23.1	25.4	26.8	26.0	24.9	22.8	17.6	13.7
		MCDB	15.8	15.3	25.2	24.9	28.2	31.8	31.8	31.2	30.9	28.0	21.0	15.8
	2%	WB	10.5	11.1	15.5	17.5	21.9	24.3	25.6	25.0	23.5	20.8	15.1	11.4
		MCDB	11.8	13.7	19.7	22.8	27.7	29.5	31.1	29.9	29.1	24.5	17.9	12.9
	5%	WB	7.9	7.9	13.4	16.1	20.8	23.3	24.7	24.1	22.4	19.2	13.6	8.3
		MCDB	9.4	9.5	16.6	21.3	26.4	28.4	29.8	28.5	27.3	22.4	16.9	9.9
	10%	WB	4.7	4.7	11.5	14.8	19.9	22.3	23.7	23.1	20.9	16.9	10.8	5.6
		MCDB	6.6	6.7	14.5	20.1	24.8	27.1	28.3	27.0	25.1	20.7	13.8	7.5
(35) Mean Daily Temperature Range	MDBR		7.7	8.2	9.4	11.5	11.0	10.7	10.9	10.8	12.2	10.8	8.7	7.0
	5% DB	MCDBR	9.7	11.5	12.9	15.1	13.1	12.8	11.9	12.2	14.4	14.1	11.6	10.0
		MCWBR	8.9	9.8	9.0	8.9	6.8	6.0	6.0	6.8	7.0	7.9	8.6	9.0
	5% WB	MCDBR	9.0	10.9	11.3	12.6	11.2	10.0	10.4	10.1	11.8	11.3	10.4	9.8
(40) Clear-Sky Solar Irradiance	taub		0.300	0.327	0.347	0.409	0.450	0.437	0.451	0.451	0.415	0.358	0.321	0.301
	taud		2.410	2.321	2.358	2.188	2.119	2.206	2.175	2.160	2.243	2.399	2.455	2.443
	Ebn at Noon		861	882	904	863	832	843	826	814	820	837	829	830
	Edh at Noon		81	103	111	141	155	142	146	144	124	95	78	73
(44) All-Sky Solar Radiation	RadAvg		1.68	2.56	3.62	4.70	5.40	6.12	6.11	5.45	4.47	3.05	2.00	1.40
	RadStd		0.17	0.26	0.28	0.43	0.49	0.41	0.35	0.31	0.35	0.28	0.23	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page